

- Gradient descent / Lagrangean methods -

Exercise 1: Gradient, Jacobian, Hessian, Step

Consider the real function of two variables:

$$f(x, y) = x^2 + 2x + y^2$$

- (1) Calculate the Jacobian,
- (2) Calculate the Hessian matrix,
- (3) Determine the characteristic equation of the Hessian matrix,
- (4) Find the eigenvalues of this matrix,
- (5) Is this matrix positive definite? Why or why not?
- (6) Calculate the determinant using the eigenvalues.
- (7) Find the optimum of this function analytically. What is its nature?
- (8) Use the gradient method to determine this extremum.
- (9) Calculate the optimal step.
- (10) Provide the algorithm corresponding to this function for calculating the step iteratively.

Exercise 2: Gradient descent algorithm-without constraints-

Let the mathematical problem $\min f(x, y)$ without constraints with:

$$f(x, y) = 4x^2 + 6y^2 + 6xy + 3x + 4y + 6$$

- (1) Calculate the minimum of f using partial derivatives.
- (2) Perform 2 iterations of the gradient descent method with fixed step $\rho = 0.1$.
- (3) Perform 2 iterations of the gradient descent method with dynamic step ρ_k .

Exercise 3: Lagrangean method-with constraints-

We want to solve the following optimization problem:

$$f(x, y) = x^2 + y^2 - 10x - 4y + 29$$

Subject to

$$y - x = -3$$

- (1) Solve the problem graphically.
- (2) Using Lagrange multipliers, find the solutions analytically.